STABLE-BASELINES3 ATARI MINI PROJECT

Chris McKie

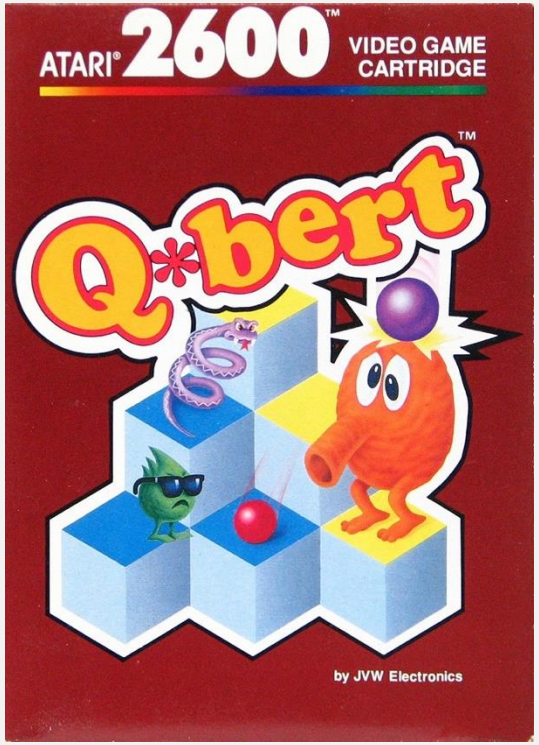
SELECTED DOMAINS & ALGORITHMS

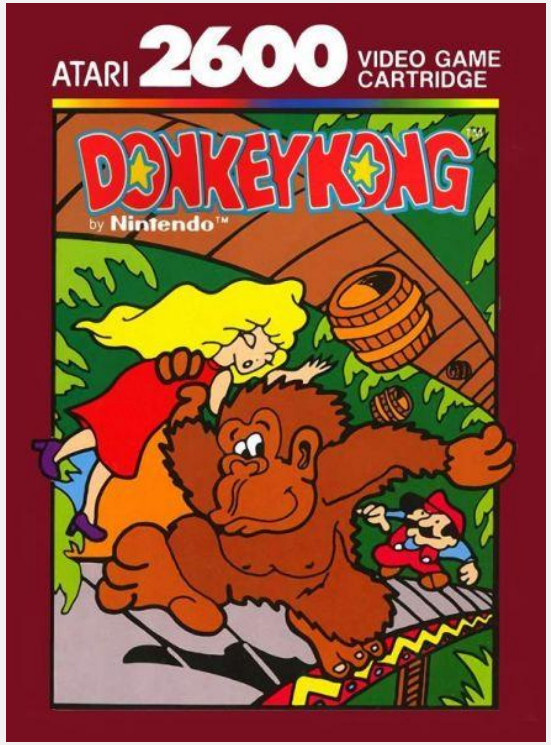
Atari Domains:

- **Q*bert:** Change the tile colors while avoiding obstacles
- **Donkey Kong:** Climb to the top of the level while avoiding barrels

Algorithms:

- Deep Q-Network (DQN)
- Proximal Policy Optimization (PPO)





HYPERPARAMETERS

Learning Rate was the primary hyperparameter variation across both domains and algorithms

<u>DQN</u>

- Learning Rate = 1e-4
- Learning Rate = 5e-5

<u>PPO:</u>

- Learning Rate = 3e-4
- Learning Rate = 1e-4

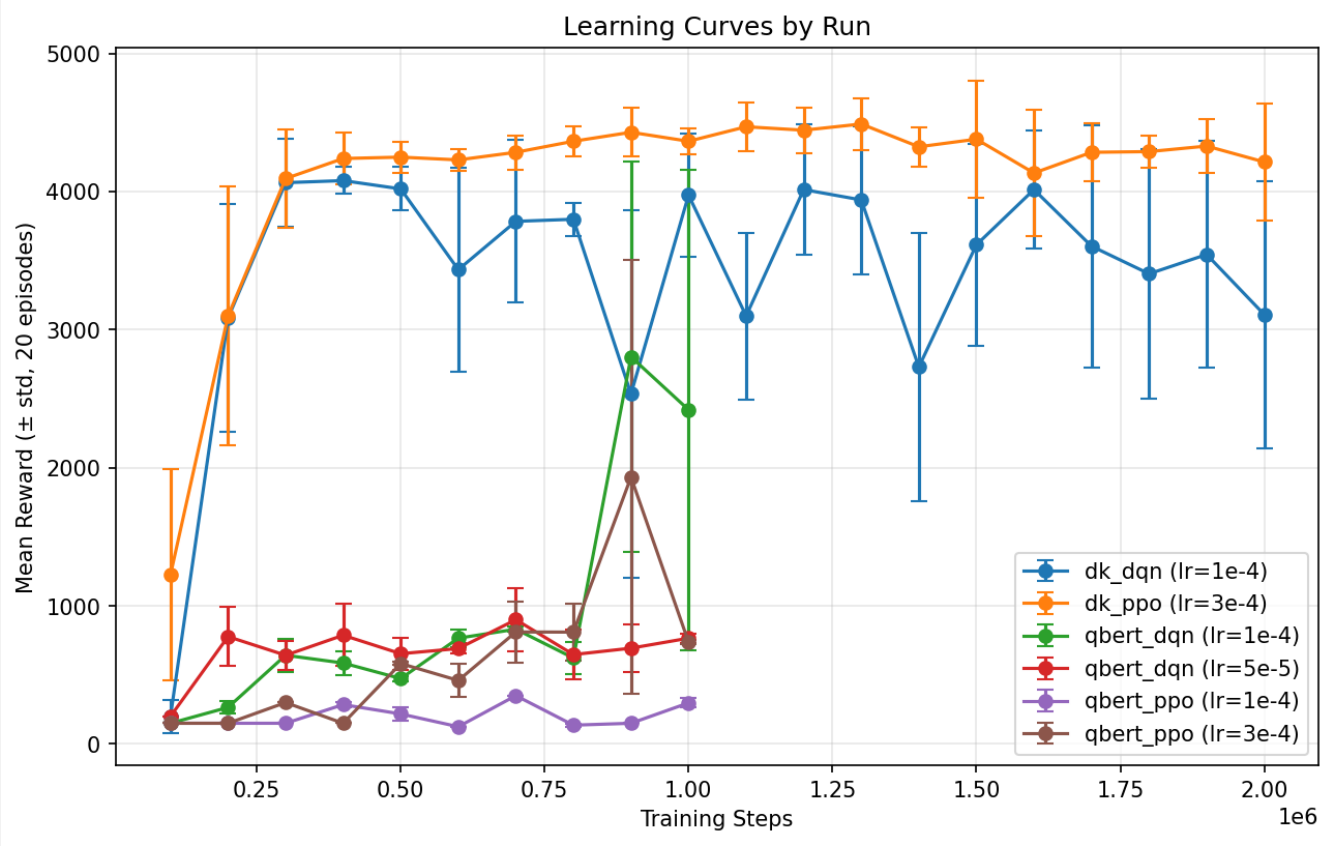
PERFORMANCE & ANALYSIS

<u>Q*bert:</u>

- Progress was slow initially
- Appears to have made new discoveries towards the end of training (especially with higher learning rates)

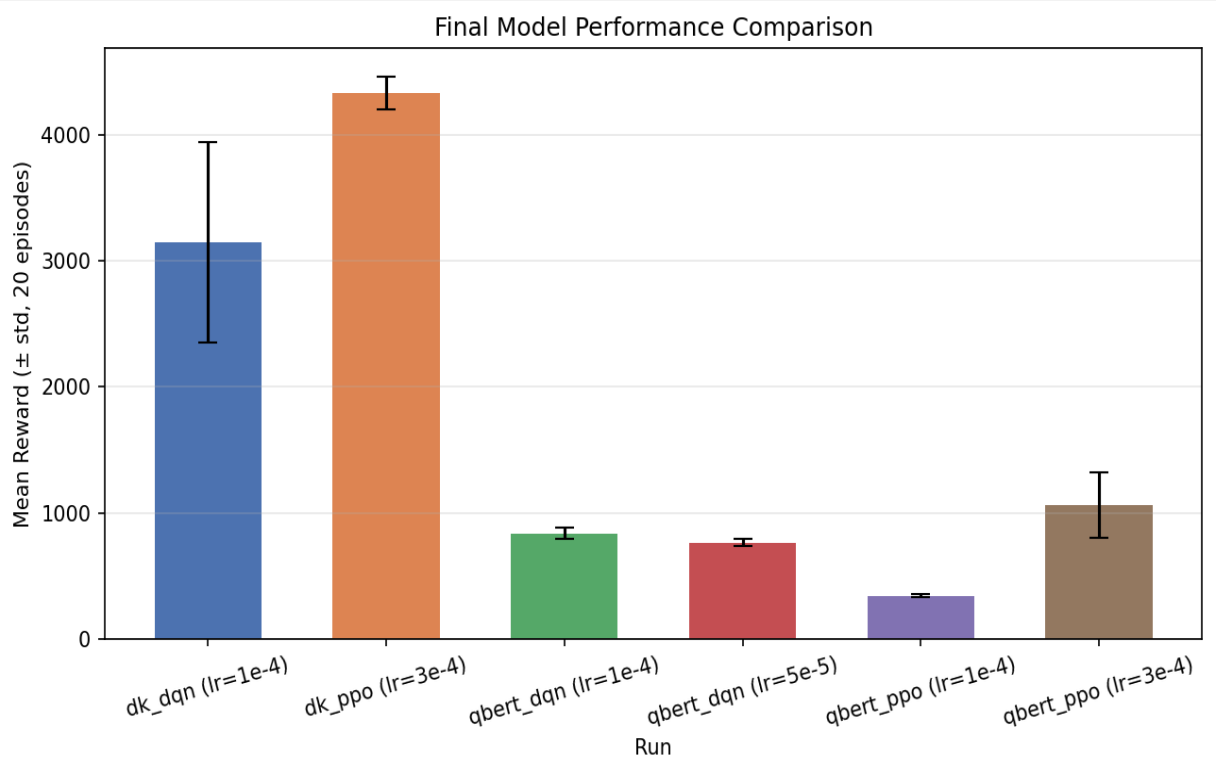
<u>Donkey Kong:</u>

- Learned to exploit jumping over barrels for points early on
- Never progressed up ladders, regardless of algorithm or parameters



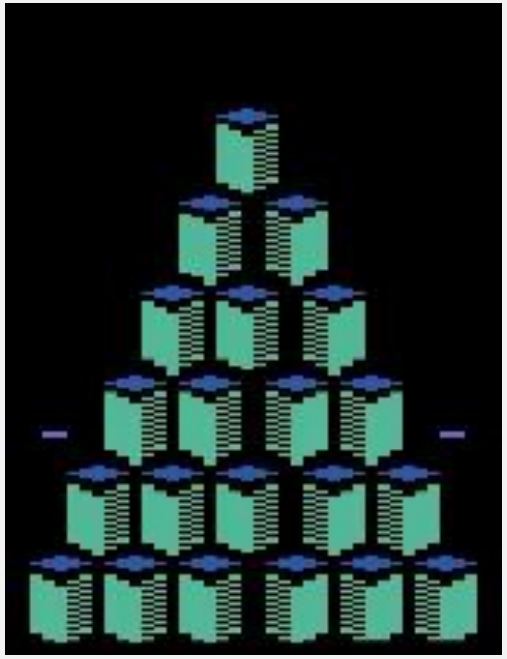
PERFORMANCE & ANALYSIS

Domain	Algorithm	Learning Rate	Mean Reward	Std. Dev.
Q*Bert	DQN	1e-4	938.8	±47.7
Q*Bert	DQN	5e-5	762.5	±29.0
Q*Bert	PPO	3e-4	1062.5	±259.0
Q*Bert	PPO	1e-4	342.5	±11.5
Donkey Kong	DQN	1e-4	3145.0	±797.2
Donkey Kong	PPO	3e-4	4330.0	±130.8

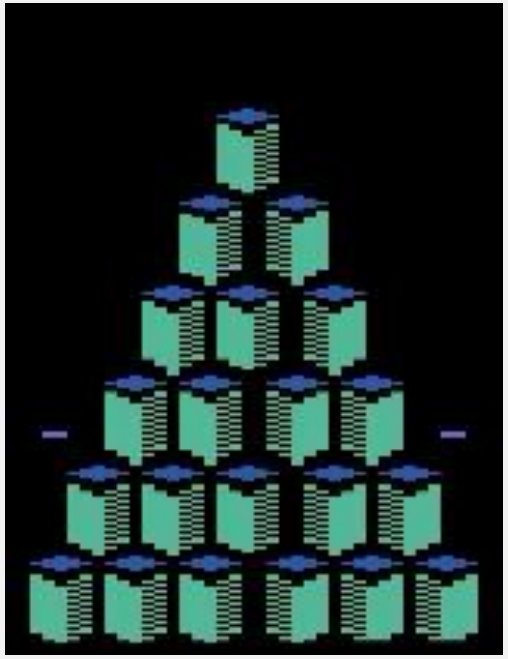


Q*BERT VIDEOS

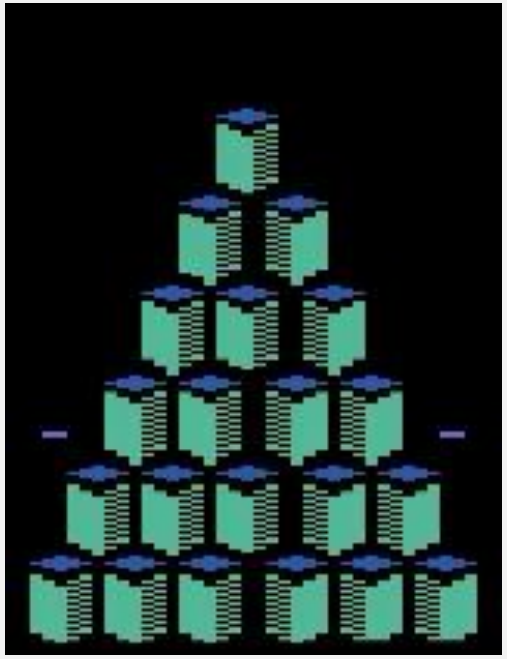
Early Model



Middle Checkpoint

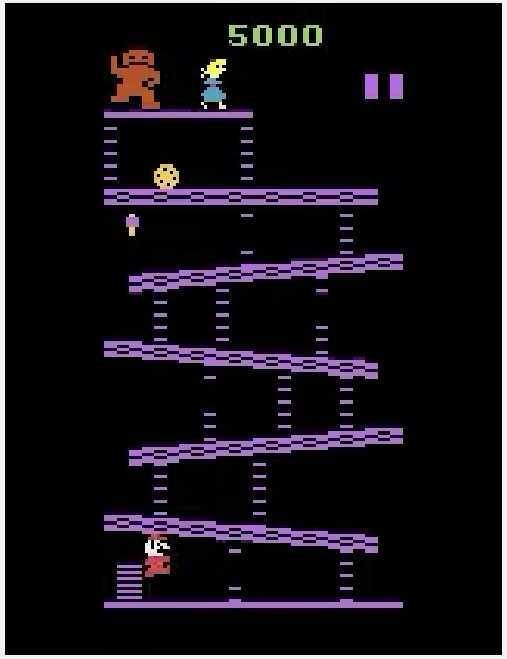


Best Model

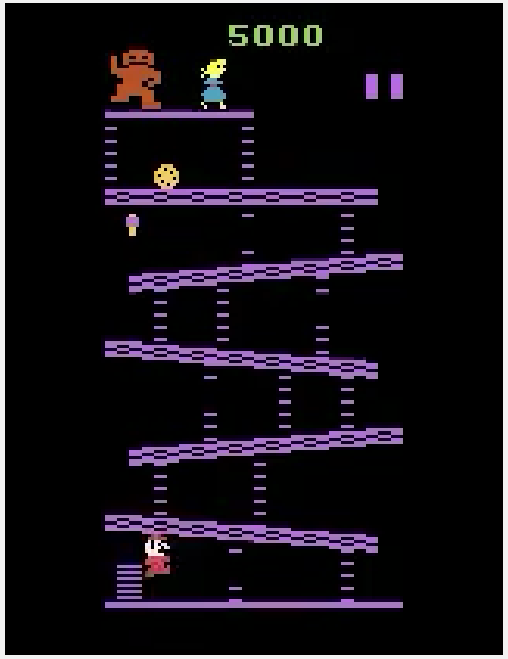


DONKEY KONG VIDEOS

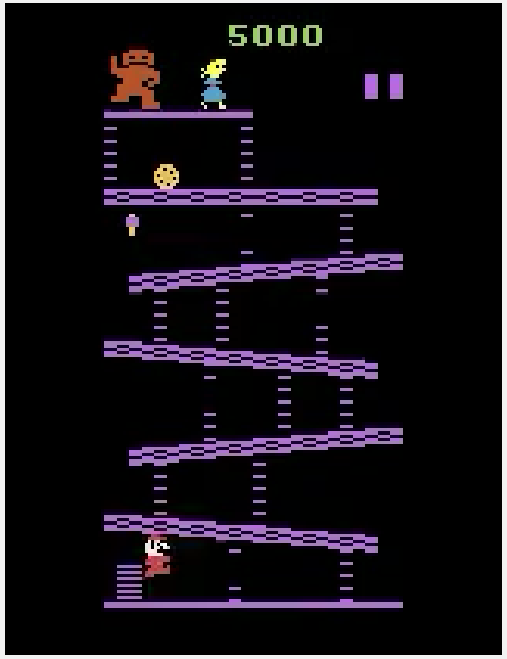
Early Model



Middle Checkpoint



Best Model



THANK YOU!